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Cohort B

MariaDB [(none)]> select @@hostname;

```
+-----+
| @@hostname |
+-----+
| Adama      |
+-----+
```

1 row in set (0.001 sec)

MariaDB [(none)]> create database ProjectSchema;

Query OK, 1 row affected (0.033 sec)

MariaDB [(none)]> use ProjectSchema;

Database changed

MariaDB [ProjectSchema]> CREATE TABLE Department (deptNo VARCHAR(10) NOT NULL
PRIMARY KEY, deptName VARCHAR(50) NOT NULL, mgrEmpNo VARCHAR(10));

MariaDB [ProjectSchema]> CREATE TABLE Employee (
-> empNo VARCHAR(10) NOT NULL PRIMARY KEY,
->
-> fName VARCHAR(30) NOT NULL,
->
-> lName VARCHAR(30) NOT NULL,
->
-> address VARCHAR(100),
->
-> DOB DATE,
->
-> sex CHAR(1) NOT NULL CHECK (sex IN ('M', 'F')),
->
-> position VARCHAR(20) NOT NULL CHECK (position IN (
->
-> 'Manager', 'Team Leader',
->
-> 'Analyst', 'Software Developer'))),
->
-> deptNo VARCHAR(10) NOT NULL,
->
-> FOREIGN KEY (deptNo) REFERENCES Department(deptNo)
->
-> ON DELETE CASCADE ON UPDATE CASCADE
->
->);

Query OK, 0 rows affected (0.040 sec)

```
MariaDB [ProjectSchema]> ALTER TABLE Department
->     ADD FOREIGN KEY (mgrEmpNo)
->         REFERENCES Employee(empNo)
->         ON DELETE SET NULL ON UPDATE CASCADE;
Query OK, 0 rows affected (0.068 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
MariaDB [ProjectSchema]> CREATE TABLE Project (
->
->     projNo    VARCHAR(10) NOT NULL PRIMARY KEY,
->
->     projName  VARCHAR(50) NOT NULL,
->
->     deptNo    VARCHAR(10),
->
->     FOREIGN KEY (deptNo) REFERENCES Department(deptNo)
->
->         ON DELETE CASCADE ON UPDATE CASCADE
-> );
Query OK, 0 rows affected (0.012 sec)
```

```
MariaDB [ProjectSchema]> CREATE TABLE WorksOn (
->     empNo      VARCHAR(10) NOT NULL REFERENCES Employee(empNo)
->         ON DELETE CASCADE ON UPDATE CASCADE,
->     projNo      VARCHAR(10) NOT NULL REFERENCES Project(projNo)
->         ON DELETE CASCADE ON UPDATE CASCADE,
->     dateWorked  DATE          NOT NULL,
->     hoursWorked DECIMAL(4,1) NOT NULL CHECK (hoursWorked BETWEEN 0 AND 40),
->     PRIMARY KEY (empNo, projNo, dateWorked)
-> );
Query OK, 0 rows affected (0.042 sec)
```

```
MariaDB [ProjectSchema]> INSERT INTO Department VALUES
->
->     ('D01', 'IT', NULL),
->
->     ('D02', 'Planning', NULL),
->
->     ('D03', 'Finance', NULL);
Query OK, 3 rows affected (0.066 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
MariaDB [ProjectSchema]> INSERT INTO Employee VALUES
```

```

->
-> ('E01', 'Alice', 'Mensah', '12 Oak St, Accra', '1985-03-14', 'F',
'Manager', 'D01'),
->
-> ('E02', 'Kwame', 'Asante', '5 Palm Ave, Tema', '1990-07-22', 'M',
'Team Leader', 'D01'),
->
-> ('E03', 'Ama', 'Boateng', '3 Ring Rd, Accra', '1992-11-05', 'F',
'Software Developer', 'D01'),
->
-> ('E04', 'Kofi', 'Owusu', '9 Hill Rd, Kumasi', '1988-01-30', 'M',
'Analyst', 'D01'),
->
-> ('E05', 'Abena', 'Darko', '20 Link Rd, Accra', '1993-06-18', 'F',
'Software Developer', 'D01'),
->
-> ('E06', 'Yaw', 'Frimpong', '7 Cedar St, Accra', '1980-09-12', 'M',
'Manager', 'D02'),
->
-> ('E07', 'Efua', 'Acquah', '2 West St, Tema', '1991-04-25', 'F',
'Team Leader', 'D02'),
->
-> ('E08', 'Nana', 'Agyei', '15 Bay Rd, Accra', '1995-02-08', 'M',
'Analyst', 'D02'),
->
-> ('E09', 'Akua', 'Bonsu', '8 North Ave, Accra', '1987-12-01', 'F',
'Manager', 'D03'),
->
-> ('E10', 'Kweku', 'Ampong', '33 South Rd, Tema', '1994-08-17', 'M',
'Software Developer', 'D03'),
->
-> ('E11', 'Adwoa', 'Sarpong', '6 East Ln, Kumasi', '1996-03-29', 'F',
'Analyst', 'D03'),
->
-> ('E12', 'Kojo', 'Mensah', '18 Cross Rd, Accra', '1991-05-14', 'M',
'Software Developer', 'D02');
Query OK, 12 rows affected (0.031 sec)
Records: 12 Duplicates: 0 Warnings: 0

```

```

MariaDB [ProjectSchema]> UPDATE Department SET mgrEmpNo = 'E01' WHERE deptNo =
'D01';
Query OK, 1 row affected (0.041 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [ProjectSchema]> UPDATE Department SET mgrEmpNo = 'E06' WHERE deptNo =
'D02';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```
MariaDB [ProjectSchema]> UPDATE Department SET mgrEmpNo = 'E09' WHERE deptNo = 'D03';
```

```
Query OK, 1 row affected (0.003 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [ProjectSchema]> INSERT INTO Project VALUES
```

```
->
->      ('P01', 'Network Upgrade', 'D01'),
->
->      ('P02', 'ERP Implementation', 'D02'),
->
->      ('P03', 'Budget System', 'D03'),
->
->      ('P04', 'Data Warehouse', 'D01'),
->
->      ('P05', 'HR Portal', 'D02');
```

```
Query OK, 5 rows affected (0.027 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [ProjectSchema]> INSERT INTO WorksOn VALUES
```

```
->
->      ('E02', 'P01', '2024-01-08', 8.0),
->
->      ('E03', 'P01', '2024-01-08', 6.5),
->
->      ('E04', 'P01', '2024-01-09', 7.0),
->
->      ('E05', 'P04', '2024-01-10', 5.5),
->
->      ('E07', 'P02', '2024-01-08', 8.0),
->
->      ('E08', 'P02', '2024-01-09', 4.0),
->
->      ('E12', 'P05', '2024-01-10', 7.5),
->
->      ('E10', 'P03', '2024-01-08', 6.0),
->
->      ('E11', 'P03', '2024-01-09', 8.0),
->
->      ('E03', 'P04', '2024-01-11', 5.0),
->
->      ('E02', 'P04', '2024-01-12', 7.0),
->
->      ('E08', 'P05', '2024-01-11', 3.5);
```

```
Query OK, 12 rows affected (0.028 sec)
```

```
Records: 12 Duplicates: 0 Warnings: 0
```

Exercise 7.25

MariaDB [ProjectSchema]> CREATE VIEW FemaleManagerProjects AS

```
-> SELECT p.projNo, p.projName, p.deptNo
-> FROM Project p
-> JOIN Department d ON p.deptNo = d.deptNo
-> JOIN Employee e ON d.mgrEmpNo = e.empNo
-> WHERE e.sex = 'F'
-> ORDER BY p.projNo;
```

Query OK, 0 rows affected (0.035 sec)

MariaDB [projectSchema]> select * from femalemanagerprojects;

projNo	projName	deptNo
P01	Network Upgrade	D01
P03	Budget System	D03
P04	Data Warehouse	D01
SCCS	Supply Chain Control System	D01

4 rows in set (0.131 sec)

Exercise 7.26

MariaDB [ProjectSchema]> CREATE VIEW EmpDetails AS

```
-> SELECT e.empNo, e.fName, e.lName, p.projName, w.hoursWorked FROM Employee e,
Project p, WorksOn w WHERE e.empNo = w.empNo AND p.projNo = w.projNo;
```

Query OK, 0 rows affected (0.037 sec)

MariaDB [projectSchema]> select * from EmpDetails;

empNo	fName	lName	projName	hoursWorked
E02	Kwame	Asante	Network Upgrade	8.0
E02	Kwame	Asante	Data Warehouse	7.0
E02	Kwame	Asante	Supply Chain Control System	6.0
E03	Ama	Boateng	Network Upgrade	6.5
E03	Ama	Boateng	Data Warehouse	5.0
E04	Kofi	Owusu	Network Upgrade	7.0
E05	Abena	Darko	Data Warehouse	5.5
E07	Efua	Acquah	ERP Implementation	8.0
E08	Nana	Agyei	ERP Implementation	4.0
E08	Nana	Agyei	HR Portal	3.5
E1	James	Ofori	Network Upgrade	5.0
E1	James	Ofori	ERP Implementation	7.0
E10	Kweku	Ampong	Budget System	6.0

E11	Adwoa	Sarpong	Budget System	8.0
E12	Kojo	Mensah	HR Portal	7.5

15 rows in set (0.034 sec)

Exercise 7.27

```
MariaDB [ProjectSchema]> CREATE VIEW EmpProject (empNo, projNo, totalHours) AS
-> SELECT w.empNo, w.projNo, SUM(hoursWorked)
-> FROM Employee e, Project p, WorksOn w
-> WHERE e.empNo = w.empNo AND p.projNo = w.projNo
-> GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.012 sec)
```

a)

```
MariaDB [ProjectSchema]> SELECT *
-> FROM EmpProject;
```

empNo	projNo	totalHours
E02	P01	8.0
E02	P04	7.0
E03	P01	6.5
E03	P04	5.0
E04	P01	7.0
E05	P04	5.5
E07	P02	8.0
E08	P02	4.0
E08	P05	3.5
E10	P03	6.0
E11	P03	8.0
E12	P05	7.5

12 rows in set (0.037 sec)

```
MariaDB [ProjectSchema]> INSERT INTO Project VALUES ('SCCS', 'Supply Chain Control
System', 'D01');
Query OK, 1 row affected (0.028 sec)
```

```
MariaDB [ProjectSchema]> INSERT INTO WorksOn VALUES ('E02', 'SCCS', '2024-01-15',
6.0);
Query OK, 1 row affected (0.027 sec)
```

```
MariaDB [ProjectSchema]> INSERT INTO Employee VALUES ('E1', 'James', 'Ofori', '22
North Rd, Accra', '1993-05-10', 'M', 'Analyst', 'D01');
Query OK, 1 row affected (0.027 sec)
```

```
MariaDB [ProjectSchema]> INSERT INTO WorksOn VALUES ('E1', 'P01', '2024-01-15', 5.0);
```

```
Query OK, 1 row affected (0.004 sec)
```

```
MariaDB [ProjectSchema]> INSERT INTO WorksOn VALUES ('E1', 'P02', '2024-01-15', 7.0);
```

```
Query OK, 1 row affected (0.027 sec)
```

b)

```
MariaDB [ProjectSchema]> SELECT projNo
```

```
-> FROM EmpProject
```

```
-> WHERE projNo = 'SCCS';
```

```
+-----+
```

```
| projNo |
```

```
+-----+
```

```
| SCCS   |
```

```
+-----+
```

```
1 row in set (0.006 sec)
```

c)

```
MariaDB [ProjectSchema]> SELECT COUNT(projNo)
```

```
-> FROM EmpProject
```

```
-> WHERE empNo = 'E1';
```

```
+-----+
```

```
| COUNT(projNo) |
```

```
+-----+
```

```
|                2 |
```

```
+-----+
```

```
1 row in set (0.002 sec)
```

d)

```
MariaDB [ProjectSchema]> SELECT empNo, totalHours
```

```
-> FROM EmpProject
```

```
-> GROUP BY empNo;
```

```
+-----+-----+
```

```
| empNo | totalHours |
```

```
+-----+-----+
```

```
| E02   |          8.0 |
```

```
| E03   |          6.5 |
```

```
| E04   |          7.0 |
```

```
| E05   |          5.5 |
```

```
| E07   |          8.0 |
```

```
| E08   |          4.0 |
```

```
| E1    |          5.0 |
```

E10	6.0
E11	8.0
E12	7.5

+-----+-----+

10 rows in set (0.003 sec)

Exercise 7.28

```
MariaDB [ProjectSchema]> create table Part(partNo VARCHAR(10), contract
VARCHAR(15), partCost double, Primary Key(partNo, contract));
Query OK, 0 rows affected (0.011 sec)
```

```
MariaDB [ProjectSchema]> INSERT INTO Part VALUES
```

```
-> ('PT01', 'CON001', 1500.00),
-> ('PT01', 'CON002', 800.00),
-> ('PT02', 'CON001', 2000.00),
-> ('PT02', 'CON003', 950.00),
-> ('PT03', 'CON002', 1200.00),
-> ('PT04', 'CON001', 750.00),
-> ('PT05', 'CON003', 3000.00),
-> ('PT06', 'CON002', 500.00),
-> ('PT07', 'CON001', 1800.00),
-> ('PT08', 'CON003', 600.00);
```

```
Query OK, 10 rows affected (0.030 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
MariaDB [ProjectSchema]> CREATE VIEW ExpensiveParts (partNo)
```

```
-> AS SELECT DISTINCT partNo
-> FROM Part
-> WHERE partCost > 1000;
```

```
Query OK, 0 rows affected (0.031 sec)
```

Discuss how you would maintain this as a materialized view and under what circumstances you would maintain the view without having to access the underlying base table Part

A materialized view is like a snapshot of a query result stored in your database, so anytime you want to access the field like the distinct partNo, you don't have to query select distinct partNo where partCost > 100 all the time, you just call the view and have access to the data. To keep the saved copy accurate, you need to update it whenever the part table changes, which you can do with triggers. The trigger would update the materialized view whenever a row is inserted, deleted, or updated from the part table. You can maintain the view without accessing the base table when inserting or updating a row where the new partCost > 1000, because the new values tells you

that the partNo qualifies to be part of the view. However, when a row is deleted or a partCost is updated downward, you would have to go back to the part table because the partNo can appear under multiple contacts, and because one contact was deleted or updated does not mean partNo should be removed from the view.